Pradyun Bachu

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Education

University of Wisconsin – Madison — Madison, WI
Bachelor of Science in Computer Science, Economics with Mathematics Emphasis — *May 2027*

- **Relevant Coursework:** *Data Structures, Machine Organization, Discrete Mathematics, Linear Algebra, Probability and Statistics, Micro/Macroeconomics*
- **Clubs and Activities:** *Claude Builder Club, Data Science Club, Wisconsin Autonomous, Federal Reserve Challenge Club, PM Club, Indian Student Association, Intramural Basketball/Volleyball/Flag Football*

Experience

Data Analyst Intern — May 2025 – August 2025
United Nations: DESA — *New York, NY*

- Engineered Python ETL pipelines leveraging Pandas for data cleaning, NumPy for statistical aggregations, and SQL for cross-dataset joins to unify **10+** global climate finance datasets spanning **65+** countries into a consistent schema, enabling reliable cross-country comparisons for senior economists
- Developed interactive dashboards visualizing **10+** key indicators, including climate finance per capita, NDC progress, and funding gaps; enabled policy teams to run side-by-side country and regional scenario comparisons
- Produced **9+** policy briefs for senior UN economists on climate finance gaps and AI's economic impact, synthesizing findings into actionable recommendations for middle-income developing nations

Event Day Judge & Logistics Team Lead — September 2020 – August 2025
PeddieHacks — *Hightstown, NJ*

- Led a **35+** member logistics team to execute a **48-hour** hackathon for **200–300+** participants annually, managing **15+** sponsor relationships and growing prize funding from **$50,000** to **$183,000**
- Served as event day judge, coordinating **12+** judges and **3+** workshop speakers, pre-screening submissions and live-judging **30+** finalist projects

Projects

Lotus — February 2026 – Present
Python, React, FastAPI, Groq API, Deepgram API, Cytoscape.js, Supabase, Vite

- Architected a healthcare cost projection platform using a two-layer detection system (regex + Groq API) to map symptoms to **40+** ICD-10 conditions with **96%** accuracy and **98.7%** F1 score; built a simulation engine on **1.5M+** MEPS patient records to model comorbidity networks and 5-year cost trajectories
- Enabled side-by-side CMS insurance plan comparison with Cytoscape.js disease progression visualization and LLM-generated recommendations to identify optimal coverage based on projected conditions and out-of-pocket costs

Voxal — December 2025 – Present
Python, React, TypeScript, FastAPI, Groq API, Deepgram API, Supabase, PostgreSQL

- Developed a voice-powered personal assistant for expense tracking, pantry management, and meal planning using Deepgram STT and Groq API, achieving **90%** classification accuracy across **22** intent types; integrated Groq Vision OCR receipt scanning at **∼560ms** avg latency handling faded, skewed, and low-contrast inputs
- Architected async FastAPI backend with Supabase Auth and PostgreSQL real-time subscriptions, enabling multi-user shared shopping lists, live pantry sync, AI spending insights, and budget alerts

Pulse — March 2026 – Present
Python, Apache Airflow, TimescaleDB, dbt, FastAPI, React, AWS (S3), Docker

- Built an end-to-end financial data pipeline using Airflow to ingest daily OHLCV data for **15** tickers (**300+** rows) into an AWS S3 landing zone and TimescaleDB hypertable across **30** days of history; dbt transforms raw data into moving averages, daily returns, z-score anomaly detection flagging **40+** unusual events, and a cross-ticker correlation matrix
- Served transformed data via FastAPI to a React dashboard with candlestick charts, anomaly markers, correlation heatmap, and inline sparklines; automated dbt data quality tests run after each DAG to validate schema and detect missing trading days

Redline — November 2025 – Present
Python, React, YOLOv8, Groq API, Flask, OpenCV

- Engineered a YOLOv8 damage detection pipeline trained on **3,000+** images across **6** classes with **3-tier** severity classification and hybrid rule-based + ML cost estimation across **20+** vehicle parts
- Built a full-stack React + Flask app with real-time damage visualization and Groq API repair chatbot; cost estimation engine factors in **8** cost components, **40+** vehicle brand valuations, and repair-vs-replace logic

Technical Skills

Programming Languages: Python, Java, SQL, C, R, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: React, FastAPI, Flask, Vite, Pandas, NumPy, Matplotlib, OpenCV
AI/ML: Groq API, Groq Vision, Deepgram API, YOLOv8
Data & Cloud Tools: Airflow, dbt, TimescaleDB, AWS (S3), Docker, Supabase, Git, Tableau